AI Guided All-Atom Protein Conformational Dynamics

MambaDiff: A Spatio-Temporal SE(3) Diffusion Framework for Protein Conformational Trajectory Modeling

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BACKGROUND

Protein function is dictated by dynamic conformational transitions, yet modeling these continuous pathways remains a challenge. While static and flow-based generative models like ConfDiff, BioEMU, and AlphaFlow effectively map conformational ensembles, they treat snapshots independently. Conversely, trajectory-based architectures such as MD-Gen, ConfRover, AlphaFolding capture dynamic paths but face computational scaling bottlenecks.

GAP IN THE FIELD

- **Static / flow-based** (ConfDiff, BioEMU, AlphaFlow): sample individual conformations, ignore temporal order.
- **Trajectory-based** (MD-Gen, ConfRover, AlphaFolding): capture motion, but scale poorly with protein size and trajectory length.
- **The gap**: no method gives real-time, real-size, time-ordered trajectories in linear complexity — until **MambaDiff**, which is **10–30× faster**.

KEY IDEA

We present **MambaDiff**, a biophysical transition model that learns macromolecular conformational dynamics directly from molecular dynamics trajectories by coupling linear-time Mamba-2 state-space architectures with $\mathrm{SE}(3)$-equivariant diffusion. Dynamical history is tracked via a **recurrent state-space model** *across time*, restricting geometric attention **locally within individual snapshots**. Mamba-2 compresses conformational memory into a *fixed-size* hidden state—scaling **linearly** with trajectory duration—while physical transition time enters as a fixed, deterministic signal.

RESULTS

	JSD ↓	Recall ↑	tICA ↑	RMSD ↓	CA%	AA%
MD (oracle)	0.31	0.74	0.18	0.00	100	99.5
AlphaFolding	0.59	0.20	N/A	3.31	10.58	0.82
MDGen	0.56	0.30	0.12	0.38	71.83	95.03
ConfRover	0.53	0.42	0.15	0.20	56.94	92.47
MambaDiff	**0.51**	**0.57**	**0.18**	**0.08**	**93.00**	**96.10**

Data — ATLAS, 1080 chains, three independent 100 ns replicas at 10 ps per frame; the 82-chain test split is held for the final evaluation.

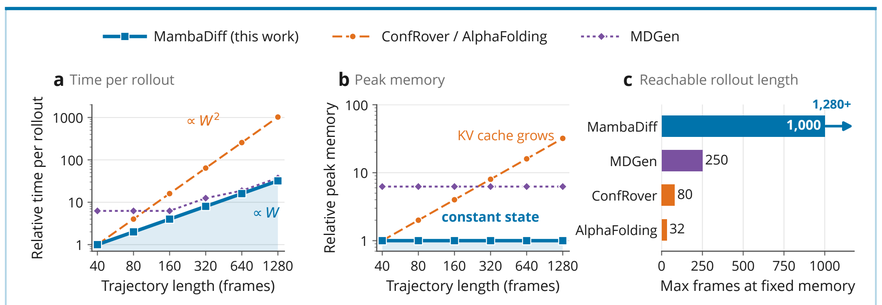


Figure 2. — MambaDiff's fixed-size recurrent state keeps time and memory per rollout step **linear** in trajectory length (**a**, **b**), where full-history attention baselines scale quadratically — reaching **1,000+ frames** at fixed memory versus 250 (MDGen), 80 (ConfRover), and 32 (AlphaFolding) (**c**).

ARCHITECTURE

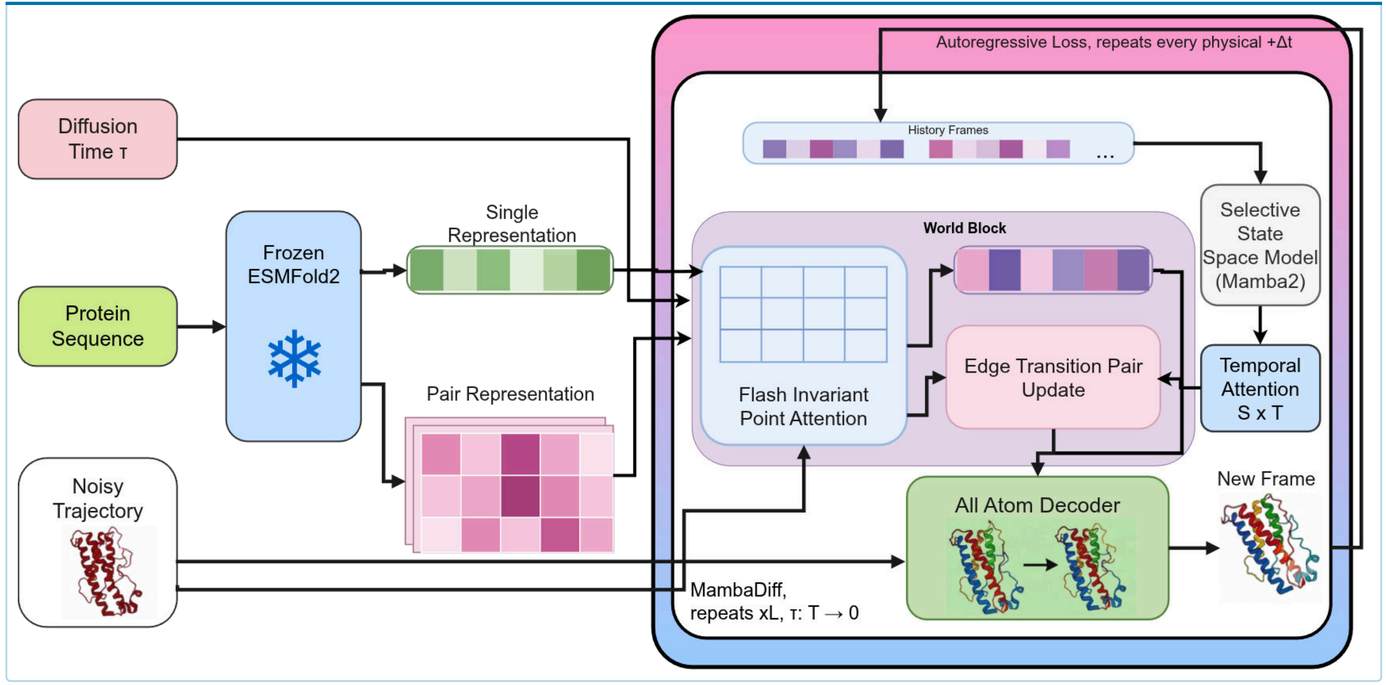


Figure 1. — **Model architecture of MambaDiff.** Protein Sequence inputs are encoded via a frozen ESMFold2 backbone to extract single and pair representations. In each diffusion step $\tau : T \to 0$ over L Layers, geometric interactions are modeled using Flash Invariant Point Attention (IPA), while temporal history is efficiently integrated using a Mamba-2 Selective State Space Model. An Edge Transition and All-Atom Update block predicts the coordinates for the next time frame, trained autoregressively across physical timesteps ($+\Delta t$).

FREE-ENERGY SURFACES AND GENERATED STRUCTURES

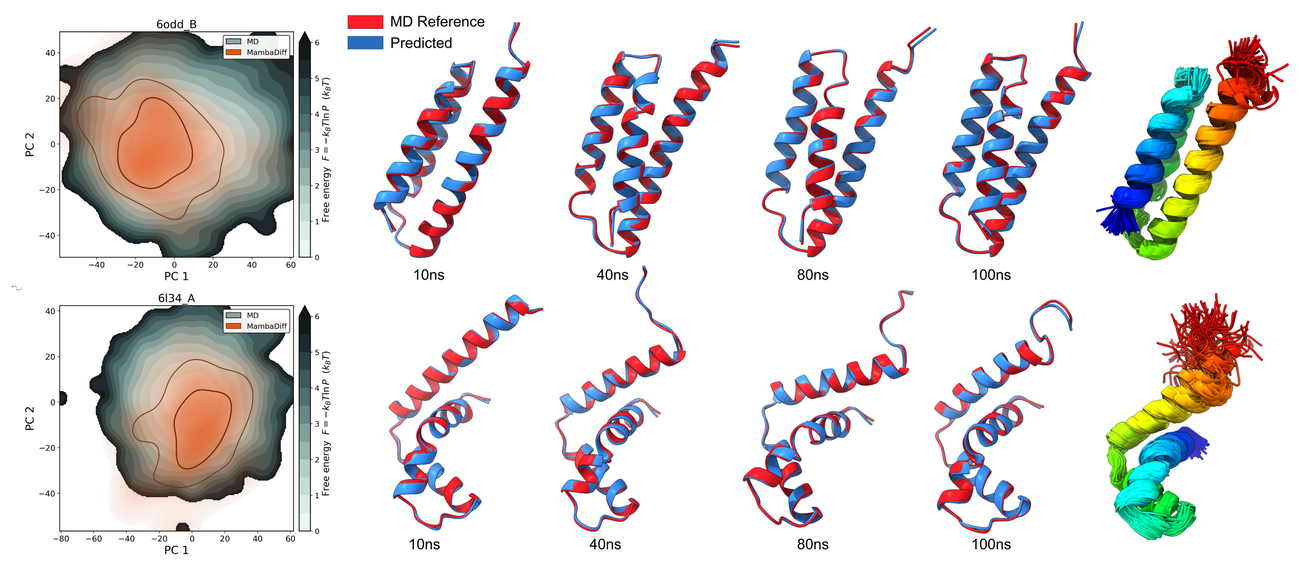


Figure 3. — Free-energy surface $F = -k_B T \ln P$ over the first two PCA components of the Cα ensemble for chains $6\mathrm{odd_B}$ and $6\mathrm{l34_A}$: reference MD (grey-blue) versus MambaDiff (orange), matched contour levels. Structures overlay MD (blue) and MambaDiff-generated (red) frames at 10, 40, and 50 ns; rightmost, the generated trajectory ensemble.

FLEXIBILITY

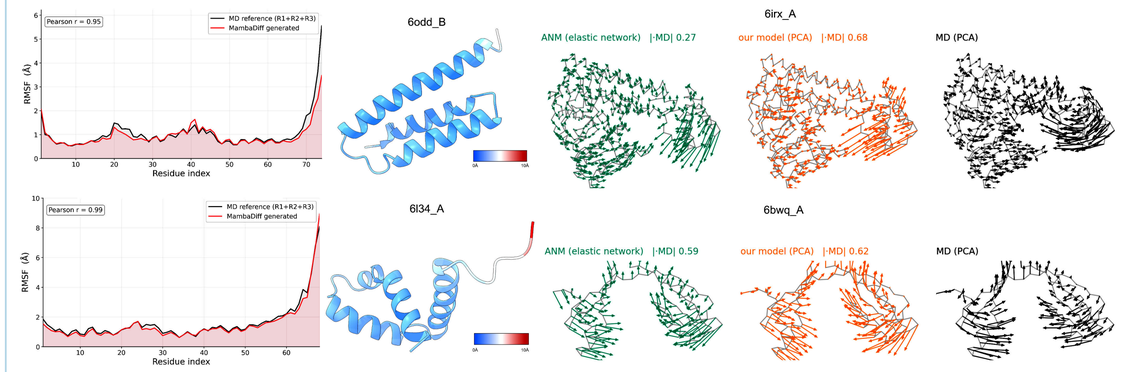


Figure 4. — **Left:** per-residue RMSF, reference MD versus MambaDiff-generated, for chains $6\mathrm{odd_B}$ and $6\mathrm{l34_A}$; the reference structure is coloured by RMSF. **Right:** dominant collective motion (first PCA mode) of $6\mathrm{irx_A}$ and $6\mathrm{bwq_A}$ from an elastic-network model (ProDy ANM), MambaDiff (PCA), and reference MD (PCA); $|\cdot\mathrm{MD}|$ gives the overlap with the MD mode.

CONFORMATIONAL COVERAGE

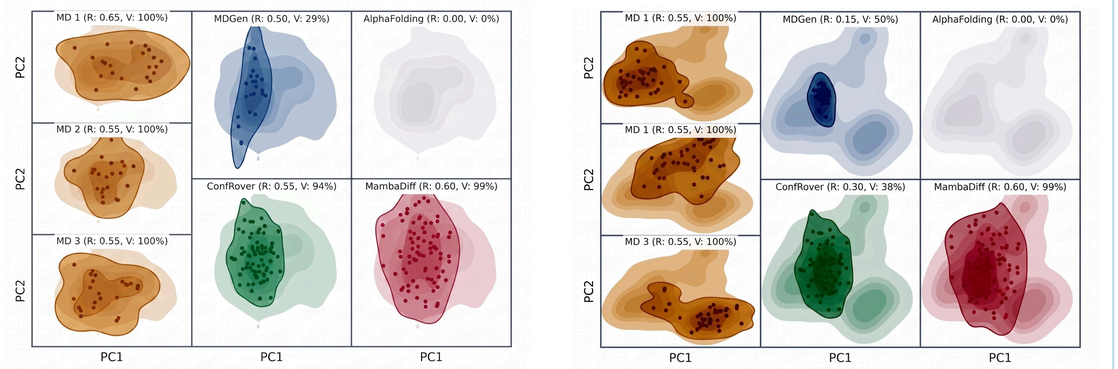


Figure 5. — PCA of the Cα ensemble for chains $6\mathrm{kty_A}$ (left) and $6\mathrm{jwh_A}$ (right); filled contours are the generated density over the reference MD landscape (grey). R is recall, V the fraction of valid conformations.

CONCLUSIONS

- SE(3)diffusion with linear-timestate-space memory generate chemically valid all-atom conformational ensembles at higher coverage than published trajectory generators.
- Trained this model on bigger **curated** dataset with different temperature and timescale for protein dynamics.
- Train on **MISATO** — $\sim$20k protein–ligand complexes with all-atom MD and QM-refined ligands — to extend the model from apo protein trajectories to bound protein–ligand simulations.

REFERENCES

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